Null Spaces and Column Spaces, $\vec{\S}4.2$

Recall from last class:

- A real **vector space** is a non-empty set V on which two operations, scalar multiplication and addition, are defined subject to the following 10 axioms. We call elements in V vectors. Let $u, v, w \in V$ and $c, d \in \mathbb{R}$.

  1. $u + v \in V$.
  2. $u + v = v + u$.
  3. $(u + v) + w = u + (v + w)$.
  4. there exists a vector 0_V in V such that $u + 0_V = u$.
  5. there exists a vector $(-u)$ in V such that $u + (-u) = 0_V$.
  6. $cu \in V$.
  7. $c(u + v) = cu + cv$.
  8. $(c + d)u = cu + du$.
  9. $c(du) = (cd)u$.
  10. $1 \cdot u = u$.

- A **subsapce** of a vector space V is a subset of V such that

  1. $0_V \in H$.
  2. H is closed under vector addition; i.e. if $u, v \in H$, then $u + v \in H$.
  3. H is closed under scalar multiplication; i.e. if $c \in \mathbb{R}$ and $u \in H$, then $cu \in H$.

- **Theorem 4.1** If $v_1, \ldots, v_p \in V$, then Span$\{v_1, \ldots, v_p\}$ is a subspace of V.

- We call Span$\{v_1, \ldots, v_p\}$ the subspace generated by or spanned by $v_1, \ldots, v_p$.

Definition 1. *The* **null space** *of an* $m \times n$ *matrix* A *is*

$$Null(A) = N(A) = \{x \in \mathbb{R}^n : A\vec{x} = \vec{0}\}.$$

Theorem: If A is an $m \times n$ matrix, then $N(A)$ is a subspace of $\mathbb{R}^n$.

1. Prove the Theorem.

2. Let $\vec{b}$ be a non-zero element in $\mathbb{R}^m$. Is $\{\vec{x} \in \mathbb{R}^n : A\vec{x} = \vec{b}\}$ a subspace of $\mathbb{R}^n$?

3. Let $A = \begin{bmatrix} 1 & -4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -5 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$. Find $N(A)$.

Definition 2. *The* **column space** *of an* $m \times n$ *matrix* A *is the span of the columns of* A. *We denote the column space of* A *by* $Col(A)$.

Theorem: If A is an $m \times n$ matrix, then $Col(A)$ is a subspace of $\mathbb{R}^m$.

4. Let $A = \begin{bmatrix} 1 & -4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -5 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$. Find Col$(A)$.

Definition 3. *Let* V *and* W *be vector spaces, then* $T : V \to W$ *is a* **linear transformation** *if*

(a) $T(u+v) = T(u) + T(v)$ *for all* $u, v \in V$ *and*

(b) $T(cu) = cT(u)$ *for all* $c \in \mathbb{R}$ *and* $u \in V$.

Definition 4. *The* **range or image** *of* T *is defined by*

$$Range(T) = im(T) = \{T(x) : x \in V\} = \{w \in W : \text{there exists } a \in V \text{ such that } T(a) = w\}.$$

The **kernel** *of* T *is defined by*

$$ker(T) = \{x \in V : T(x) = 0_W\}.$$

5. Define $T : \mathbb{P}_n \to \mathbb{P}_{n-1}$ by $T(f) = \frac{df}{dt}$.

(a) Is T linear?

(b) Find the image of T.

(c) Find the kernel of T.

6. Define $T : \mathbb{P}_2 \to \mathbb{P}_3$ by $T(p) = tp(t) + p'(t)$.

(a) Find the range of T.

(b) Find the kernel of T.

(c) Is T onto? Is T one-to-one?

7. Let $T : \mathbb{P}_2 \to \mathbb{R}^2$ by $T(p) = \begin{bmatrix} p(0) \\ p(1) \end{bmatrix}$.

(a) Is T linear?

(b) Find $\ker(T)$ and $\mathrm{Range}(T)$.

8. Define $T : \mathbb{P}_2 \to M_{2\times 2}$ by $T(p) = \begin{bmatrix} p(1) - p(0) & 0 \\ 0 & p(0) \end{bmatrix}$. Find the kernel and image of T.